



Achieving Dimension-Free Communication in Federated Learning via Zeroth-Order Optimization



Zhe Li^{1*}, Bicheng Ying^{2*}, Zidong Liu³, Chaosheng Dong⁴, Haibo Yang¹

¹Rochester Institute of Technology, ²Google, ³ComboCurve, ⁴Amazon



★ Motivation & Contribution

- Huge communication costs (TB level) linearly scaling with model dimension in FL, especially in LLM tasks.
- Propose **DeComFL**, achieving the MB level communication costs in LLM fine-tuning via ZOO.

★ Zeroth-Order Optimization (ZOO)

- Estimate gradient by function value difference
- ZO gradient scalar:
$$g = \frac{f(\mathbf{x} + \mu \mathbf{z}) - f(\mathbf{x})}{\mu}, \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$
- ZO-SGD: $\mathbf{x}_{r+1} = \mathbf{x}_r - \eta g \mathbf{z}$
- Model update term “gz” consists of:
 - 1) scalar g
 - 2) perturbation vector $\mathbf{z}$ (can be generated by a fixed random seed - a scalar as well)

★ Communication Complexity

Table 1: Comparison of Total Communication Complexity of Typical Algorithms

Algorithm	Uplink Comm. Per Round	Downlink Comm. Per Round	Round Complexity	Uplink Comm. Complexity	Downlink Comm. Complexity
FedAvg	md	md	$\mathcal{O}(1/mK\epsilon^2)$	$\mathcal{O}(d/K\epsilon^2)$	$\mathcal{O}(d/K\epsilon^2)$
FedZO	md	md	$\mathcal{O}(d/mPK\epsilon^2)$	$\mathcal{O}(d^2/PK\epsilon^2)$	$\mathcal{O}(d^2/PK\epsilon^2)$
FedCom	βmd	md	$\mathcal{O}(1/mK\epsilon^2)$	$\mathcal{O}(\beta d/K\epsilon^2)$	$\mathcal{O}(d/K\epsilon^2)$
Ours (Standard)	mKP	$2MKP$	$\mathcal{O}(d/mPK\epsilon^2)$	$\mathcal{O}(d/\epsilon^2)$	$\mathcal{O}(Md/m\epsilon^2)$
Ours (Low rank)	mP	$2MP$	$\mathcal{O}(\kappa/mP\epsilon^2)$	$\mathcal{O}(\kappa/\epsilon^2)$	$\mathcal{O}(M\kappa/m\epsilon^2)$

★ DeComFL Algorithm Design

Comm. cost is independent of the model dimension!

Algorithm 1 Dimension-Free Communication in Federated Learning (DeComFL)

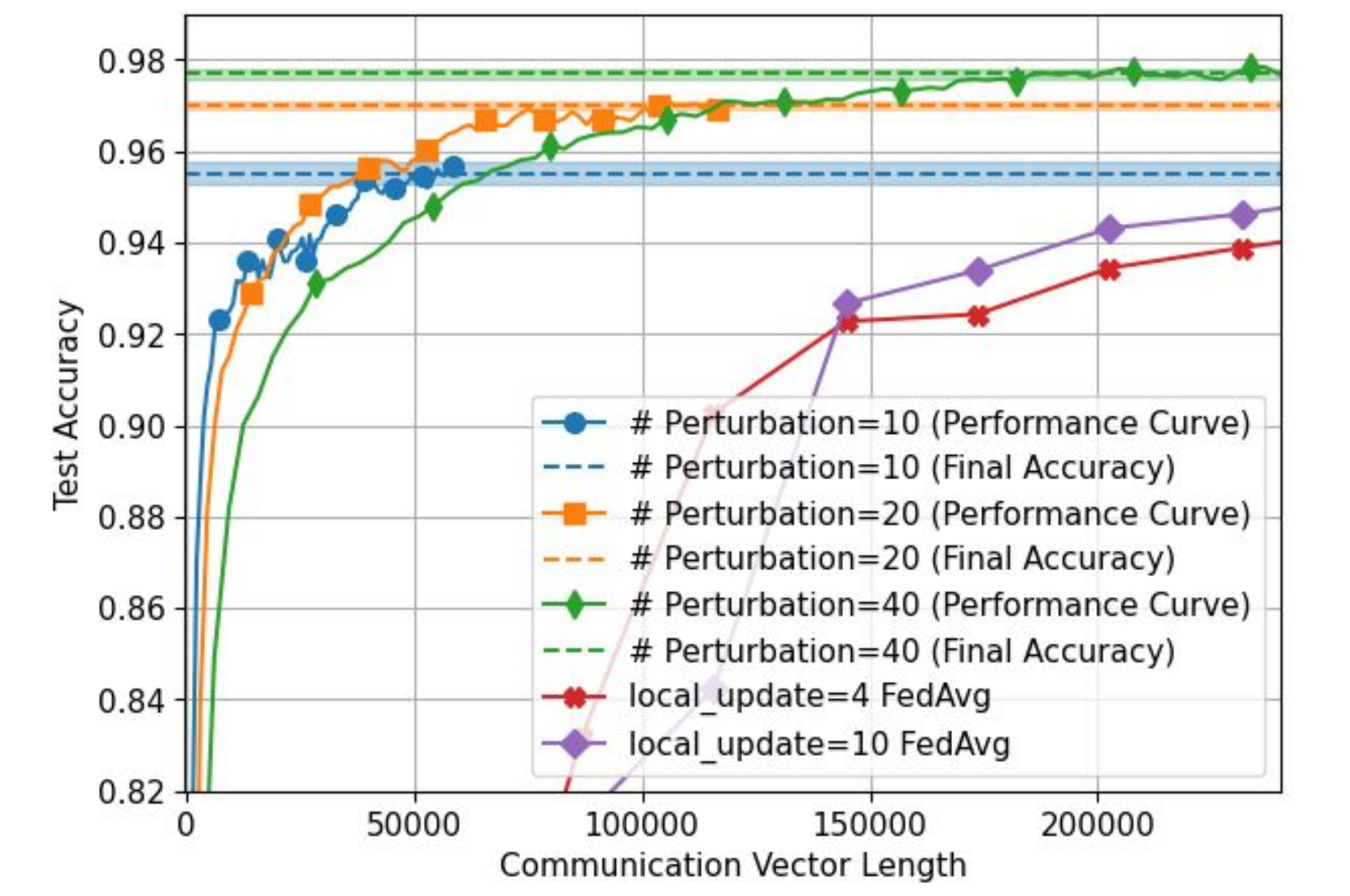
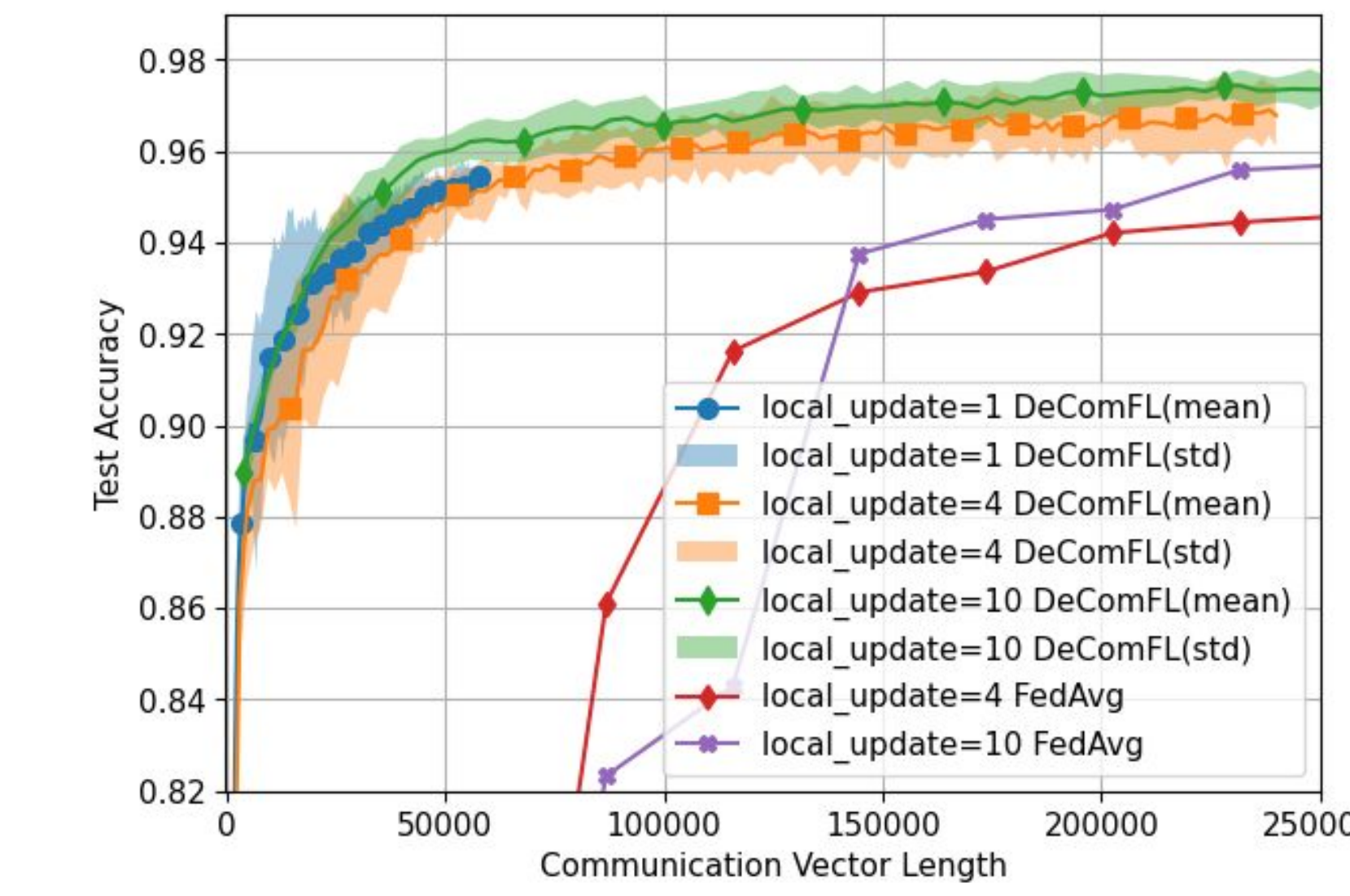
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1: Initialize:  $\{g_0^k\}_{k=1}^K$ , learning rate  $\eta$ , local update steps  $K$ , communication rounds  $R$ .
2: Allocate: memory for recording three states: 1) state set  $\{t_i\}_{i=1}^N$  storing the last round that client  $i$  participated in, 2) seed set  $\{\{s_r^k\}_{k=1}^K\}_{r=0}^{R-1}$ , 3) gradient set  $\{\{g_r^k\}_{k=1}^K\}_{r=0}^{R-1}$ .
3:
4: for  $r = 0, 1, \dots, R - 1$  do
5:   Uniformly sample a client set  $C_r$  with cardinality  $m$  and sample  $K$  seeds  $\{s_r^k\}_{k=1}^K$ 
6:   for each client  $i \in C_r$  in parallel do
7:     ClientRebuildModel( $\{\{g_{r'}^k\}_{k=1}^K\}_{r'=t_i}^{r-1}, \{\{s_{r'}^k\}_{k=1}^K\}_{r'=t_i}^{r-1}$ )  $\triangleright$  Send  $g$  and  $s$  to client
8:      $\{g_{i,r}^k\}_{k=1}^K = \text{ClientZOLocalUpdate}(\{s_r^k\}_{k=1}^K, r)$   $\triangleright$  Send  $s$  to client and receive  $g$ 
9:   end for
10:  Compute the global gradient scalars  $\{g_r^k\}_{k=1}^K = \left\{ \frac{1}{|C_r|} \sum_{i \in C_r} g_{i,r}^k \right\}_{k=1}^K$ 
11:  Store  $\{g_r^k\}_{k=1}^K$  and  $\{s_r^k\}_{k=1}^K$  and update the client's last update record  $t_i = r$ 
12:   $\mathbf{x}_{r+1} = \mathbf{x}_r - \eta \sum_{k=1}^K g_r^k \cdot \mathbf{z}_r^k$   $\triangleright$  This step is optional
13: end for
14:
15: procedure 1. ClientRebuildModel( $\{\{g_{r'}^k\}_{k=1}^K\}_{r'=t_i}^{r-1}, \{\{s_{r'}^k\}_{k=1}^K\}_{r'=t_i}^{r-1}$ )
16:   for  $r' = t_i, \dots, r - 1$  do
17:     for  $k = 1, \dots, K$  do
18:       Generate  $\mathbf{z}_{r'}^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$  by random seed  $s_{r'}^k$ .
19:        $\mathbf{x}_{i,r'}^{k+1} = \mathbf{x}_{i,r'}^k - \eta \cdot g_{r'}^k \cdot \mathbf{z}_{r'}^k$   $\triangleright \mathbf{x}_{i,t_i}^0$  is the local model
20:     end for
21:   end for
22: end procedure
23:
24: procedure 2. ClientZOLocalUpdate( $\{s_r^k\}_{k=1}^K, r$ )  $\triangleright$  Can be replaced by other ZO methods
25:   for  $k = 1, \dots, K$  do
26:     Generate  $\mathbf{z}_r^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$  by random seed  $s_r^k$ 
27:      $g_{i,r}^k = \frac{1}{\mu} (f_i(\mathbf{x}_{i,r}^k + \mu \mathbf{z}_r^k; \xi_{i,r}^k) - f_i(\mathbf{x}_{i,r}^k; \xi_{i,r}^k))$   $\triangleright$  Forward difference method
28:      $\mathbf{x}_{i,r}^{k+1} = \mathbf{x}_{i,r}^k - \eta \cdot g_{i,r}^k \cdot \mathbf{z}_r^k$   $\triangleright$  Standard ZO-SGD
29:   end for
30:   revert the local model back to  $\mathbf{x}_{i,r}^1$ .  $\triangleright$  This step is crucial
31:   return  $\{g_{i,r}^k\}_{k=1}^K$ 
32: end procedure

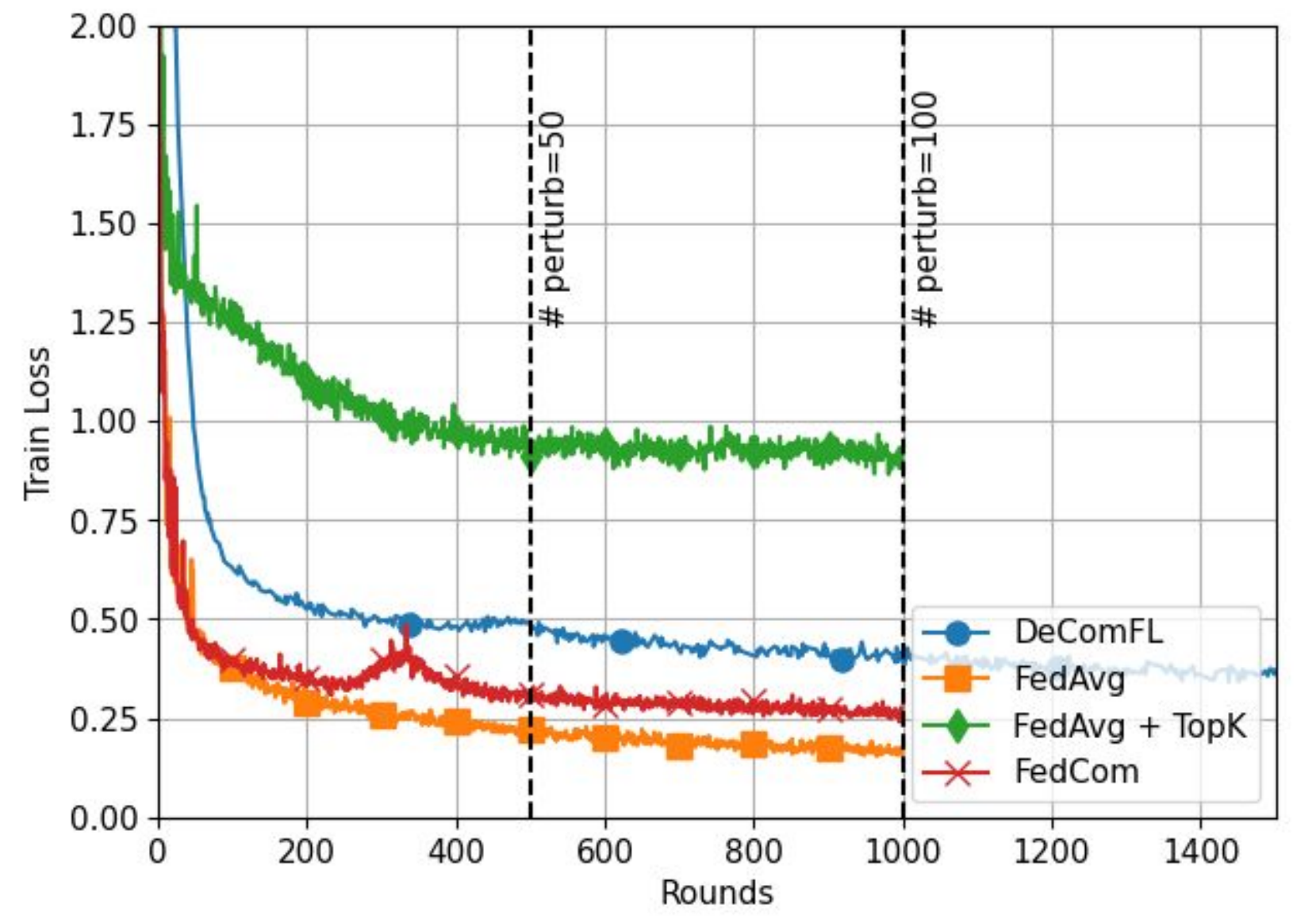
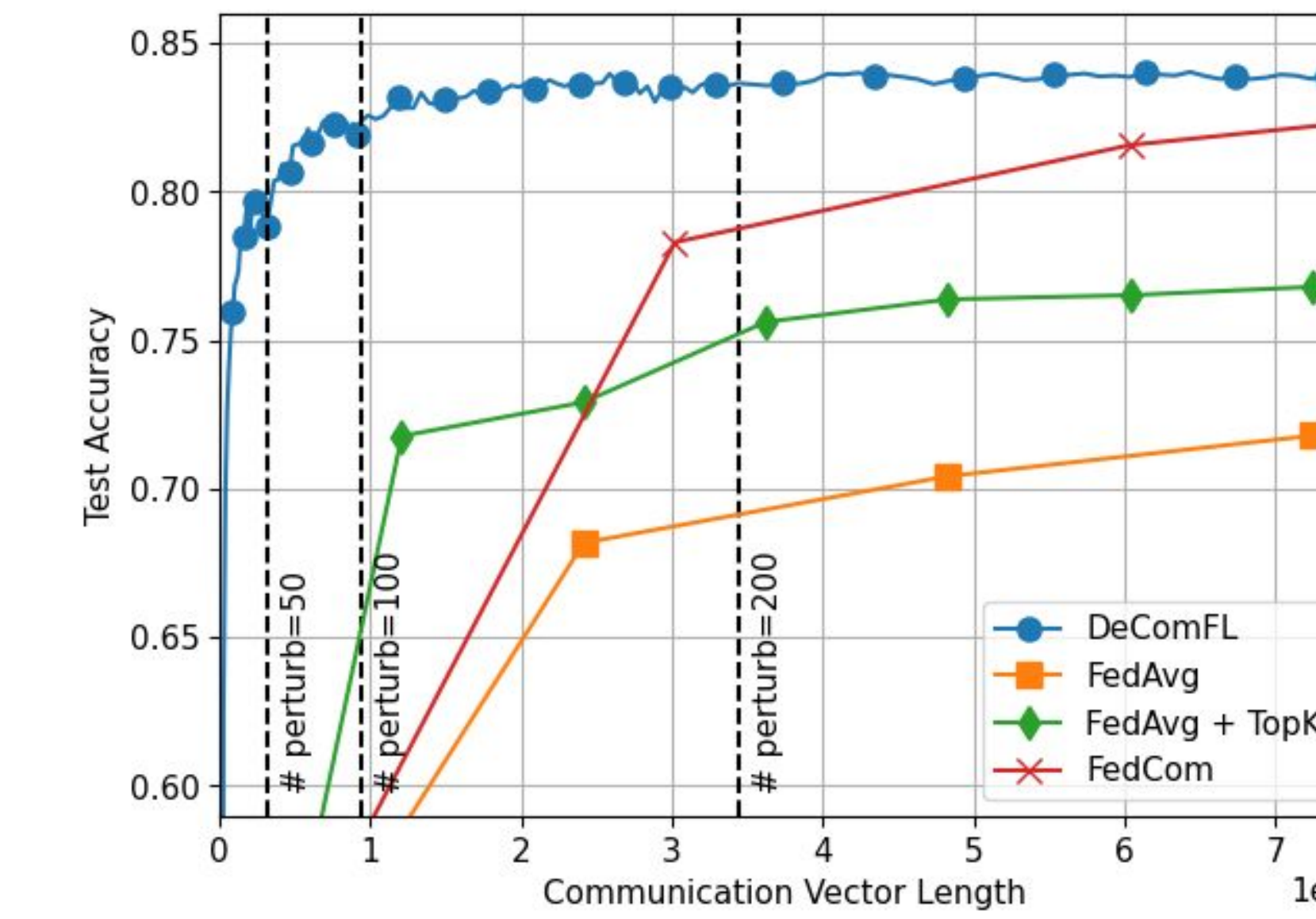
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★ Experiments

- Training Setup: 100 clients. 10% clients participate.
- Ablation study of local updates and perturbations:



- Comparison of Multiple FL Algorithms on Fashion:



- Fine-tuning Setup: 8 clients. 2 clients participate.

Table 2: Test Accuracy and Communication Cost on Fine-Tuning Tasks

Model	Dataset \ Task	FedAvg	MeZO	FedZO with $P = 10$	DeComFL with $P = 10$
OPT-125M	SST-2	87.32% (0.24 TB)	83.99%	84.11% (0.75 TB)	85.08% (0.36 MB)
	CB	82.14% (0.12 TB)	72.49%	74.41% (0.38 TB)	75.00% (0.12 MB)
	WSC	63.25% (0.12 TB)	55.18%	59.47% (0.75 TB)	59.59% (0.36 MB)
	WIC	60.83% (0.12 TB)	53.25%	53.37% (0.75 TB)	53.38% (0.36 MB)
	RTE	63.96% (0.48 TB)	52.91%	54.16% (0.50 TB)	57.05% (0.24 MB)
	BoolQ	62.34% (0.24 TB)	61.46%	61.25% (0.50 TB)	61.60% (0.24 MB)
OPT-1.3B	SST-2	90.38% (1.27 TB)	90.23%	90.33% (5.20 TB)	90.78% (0.24 MB)
	CB	83.93% (1.27 TB)	74.01%	74.49% (7.80 TB)	75.71% (0.36 MB)
	WSC	65.65% (1.27 TB)	58.21%	61.11% (7.80 TB)	64.16% (0.36 MB)
	WIC	65.82% (1.27 TB)	55.95%	56.08% (5.20 TB)	56.14% (0.24 MB)
	RTE	66.13% (2.54 TB)	57.57%	59.21% (3.90 TB)	60.89% (1.80 MB)
	BoolQ	63.83% (5.08 TB)	61.98%	62.14% (3.90 TB)	62.50% (1.80 MB)